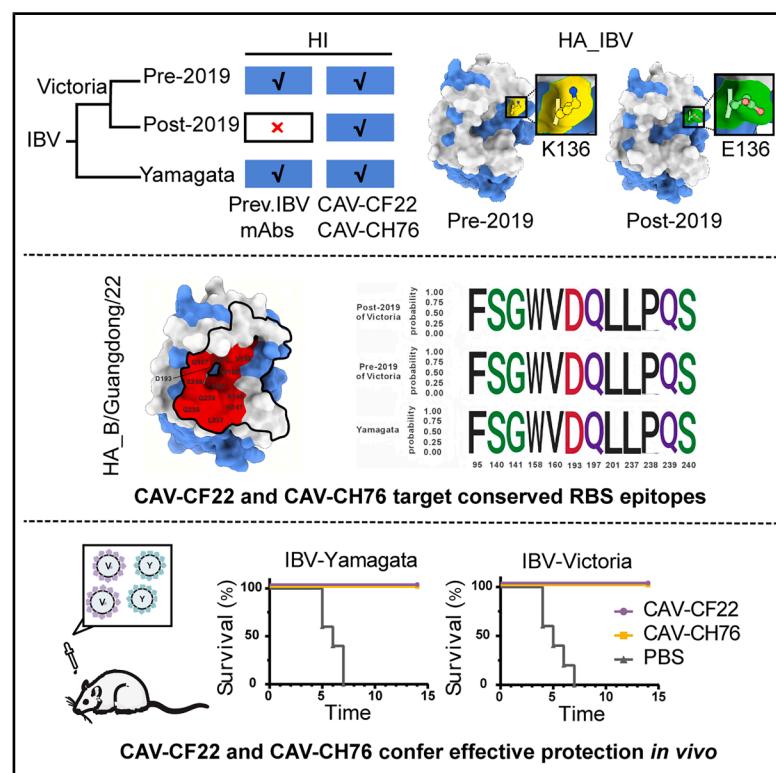


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Human monoclonal antibodies isolated after seasonal vaccination broadly neutralize antigenically drifted influenza B viruses

Graphical abstract



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In brief

Cai et al. report two human monoclonal antibodies, CAV-CF22 and CAV-CH76, that neutralize post-2019 antigenically drifted influenza B viruses. Structural studies demonstrate a highly conserved binding epitope in the HA head, supporting vaccine and antibody strategies that are resilient to recent IBV drift.

Highlights

- CAV-CF22 and CAV-CH76 broadly neutralize Victoria- and Yamagata-lineage IBVs
- Post-2019 K136E substitution undermines many older head bnAbs yet spares CAV-CF22/-CH76
- Both antibodies recognize a highly conserved epitope in the RBS region of IBV HA
- Both mAbs confer broad, potent prophylactic and therapeutic protection against IBV

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Article

Human monoclonal antibodies isolated after seasonal vaccination broadly neutralize antigenically drifted influenza B viruses

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SUMMARY

Antigenic drift in hemagglutinin (HA) enables influenza viruses to escape host immunity. Elucidating molecular features of antigenic drift is essential for updating seasonal vaccines and pandemic preparedness. Here, we found that influenza B viruses (IBVs) isolated after 2019 escaped neutralization by several previously identified broadly neutralizing monoclonal antibodies (bnAbs). Meanwhile, we identified two IBV bnAbs, CAV-CF22 and CAV-CH76, isolated via quadrivalent vaccine. They exhibited broad neutralizing activity against Victoria- and Yamagata-lineage viruses *in vitro* and protected *in vivo* against contemporary Victoria and Yamagata strains. Phylogenetic and structural analysis revealed fixation of K136E in post-2019 Victoria HA, disrupting epitopes targeted by most previously characterized head-directed IBV-monoclonal antibodies (mAbs). High-resolution structures reveal that, rather than engaging K136, CAV-CF22 and CAV-CH76 insert HCDR3 into the receptor-binding site (RBS) to sterically mimic sialic acid. The conservation of epitope residues underlies the antibodies' broad neutralizing activity against IBV and informs antibody- and vaccine-design strategies that are resilient to recent IBV drift.

INTRODUCTION

Influenza viruses cause seasonal respiratory disease, with annual epidemics driven largely by the co-circulation of influenza A virus (IAV) and influenza B virus (IBV).¹ These epidemics typically infect 5%–15% of the global population each year.^{2,3} Influenza causes approximately 650,000 deaths each year, with the IBV contributing to 20% to 30% of these fatalities.⁴ During some flu seasons, the prevalence of IBV dominates, leading to a large number of hospitalizations.^{5,6} Between 2003 and 2008, IBV spread widely throughout China, causing more than half of all influenza-associated deaths.⁷ IBV disproportionately affects children and the elderly, accounting for up to 52% of pediatric

influenza-related fatalities.⁸ Current therapeutic options for IBV include neuraminidase inhibitors, hemagglutinin (HA)-targeting agents, and viral polymerase inhibitors, but extensive antiviral use has promoted the emergence of resistant strains, and clinical effectiveness against IBV remains limited.^{9,10} Together, these factors underscore the persistent need for improved, broadly protective strategies for the prevention and treatment of IBV.

HA mediates receptor attachment and is the principal target of neutralizing antibodies. Most broadly, head-directed antibodies typically recognize the receptor-binding site (RBS) or adjacent conserved elements. While RBS-focused antibodies are well documented for influenza A subtypes (e.g., H1N1 and H3N2),

only a handful of head-directed broadly neutralizing IBV antibodies (CR8033, C12G6, H1209, H1272, and H2365) have been described to date.^{11–13} According to the report from the World Influenza Center (<https://www.crick.ac.uk/research/platforms-and-facilities/worldwide-influenza-centre/annual-and-interim-reports>), the serum from ferrets infected with B/Colorado/06/2017 (2019 and 2020 vaccine strain) shows significantly weakened hemagglutination inhibition (HI) activity against Victoria lineage viruses isolated after 2019. Recent surveillance has documented accumulating substitutions and short deletions in antigenic sites of the IBV-HA head, raising concern that previously protective antibodies may no longer be effective.

Antigenic evolution of IBV has involved amino acid substitutions and short deletions in key antigenic loops. Viruses from the 2017 and 2018 seasons carried a two amino acid deletion at HA1 positions 162–163 in the 160-loop, a change represented by the vaccine strain B/Colorado/06/2017.^{14,15} Since late 2019, the Victoria lineage has dominated global circulation, while B/Yamagata has not been detected in the wild since March 2020.^{16–18} According to the World Influenza Centre's report, post-2019 Victoria viruses have acquired additional substitutions at HA1 positions D129, G133, K136, N150, and K203, several of which map to or near the RBS. Given that even single changes in or adjacent to the RBS can impair seasonal vaccine effectiveness, it is critical to determine whether seasonal vaccines can induce humoral responses capable of countering antigenic evolution in post-2019 IBV strains.

Here, through antigenic cartography analysis of ferret antisera, we found that IBV strains emerging after the second half of 2019 underwent antigenic drift, resulting in immune escape (<https://www.crick.ac.uk/research/platforms-and-facilities/worldwide-influenza-centre/annual-and-interim-reports>). We demonstrated that previously identified broadly neutralizing IBV-monoclonal antibodies (mAbs) cannot neutralize post-2019 Victoria-lineage viruses. Moreover, we reported two human mAbs (CAV-CF22 and CAV-CH76) targeting the RBS of IBV HA. Both mAbs demonstrated potent cross-lineage neutralization against historically circulating Yamagata and contemporary Victoria lineages of IBV and conferred robust prophylactic and therapeutic protection *in vivo* against lethal IBV challenge. Structural analysis shows that a K136E substitution, characteristic of post-2019 Victoria-lineage viruses, disrupts binding of most previously characterized HA-head mAbs. By contrast, CAV-CF22 and CAV-CH76 exhibit minimal or no dependence on residue K136. High-resolution structures reveal a conserved aspartic acid at the tip of the HCDR3 (D105 in CAV-CH76 and D107 in CAV-CF22) that appears central to their mode of RBS engagement. These results clarify the molecular basis for recent escape from many reported IBV head-directed antibodies and identify an RBS-centered vulnerability that can inform the design of vaccines and antibody therapeutics with increased resilience to ongoing antigenic drift.

RESULTS

Previously reported broadly neutralizing HA head-directed mAbs fail to neutralize post-2019 IBV strains

BnAbs targeting the IBV-HA head domain typically engage the RBS. Previously reported bnAbs, including CR8033, C12G6, H1209, H1272, and H2365, bind to this conserved RBS region.

Previous reports from the World Influenza Center indicated that the B/Victoria-lineage vaccine strain for 2019 and 2020 showed weakened immune responses against viruses isolated after 2019 (<https://www.crick.ac.uk/research/platforms-and-facilities/worldwide-influenza-centre/annual-and-interim-reports>). The V1A.3 strain of the virus that was prevalent in 2019 may affect the effectiveness of previous vaccine strains (<https://nextstrain.org/>). To investigate whether the previously reported broadly neutralizing antibodies remain effective against the currently circulating IBV strains, we tested the binding of H1209, H1272, H2365, C12G6, CR8033, and the control antibody CR8059 against 14 IBV strains from different lineages via enzyme-linked immunosorbent assay (ELISA) (Figure 1A). Binding to B/Victoria-lineage strains isolated after 2019 was significantly reduced. We next compared the *in vitro* HI and neutralization activities, as well as the breadth of reactivity (Figures 1B and 1C). Consistent with the antibody binding results, all of these antibodies lost their HI activity and neutralizing capacity against post-2019 IBV strains. C12G6 and CR8033 lacked HI activity and neutralizing effect against the B/Colorado/06/2017 (V). H1209, H1272, and H2365 displayed strong HI activity and neutralizing capacity against the tested B/Colorado/06/2017 (V) and six additional strains.

Influenza vaccination can induce broadly neutralizing antibodies in humans

To explore whether influenza vaccination can effectively induce a humoral immune response to overcome the antigenic drift of IBV, we isolated B cells from influenza-vaccinated volunteers. Peripheral blood mononuclear cells (PBMCs) were collected from donors immunized with the 2020 and 2021 seasonal quadrivalent influenza virus split vaccine containing the following strains: A/Guangdong-Maonan/SWL1536/2019 (H1N1) pdm09-like virus, A/Hongkong/2671/2019 (H3N2)-like virus, B/Washington/02/2019-like virus (Victoria lineage), and B/Phuket/3073/2013-like virus (Yamagata lineage). Vaccine-induced plasmablasts (PBs) were sorted via flow cytometry into single-cell PCR plates.^{19,20} The immunoglobulin variable genes of B cells were amplified using reverse transcription polymerase chain reaction (RT-PCR), sequenced, and subsequently cloned into human IgG1 expression vectors for expression in mammalian cells (Figure 2A).

To investigate the functions of the obtained antibodies, first, we used ELISA to evaluate the binding capacity of these isolated antibodies to the HA of IBV. We involved three previously reported IBV antibodies (C12G6, CR8033, and CR8059) for comparison. Both C12G6 and CR8033 target the HA head, while CR8059 binds on the vestigial esterase domain at the base of the HA head. C12G6 and CR8033 showed weak binding capacity to the HA of B/Colorado/06/2017 (V), while CR8059 exhibited cross-lineage binding to the HA of both IBV lineages (Figure 2B). Notably, seven of our isolated antibodies demonstrated cross-lineage binding to HA antigens from Victoria (B/Colorado/06/2017) and Yamagata (B/Florida/4/2006) strains (Figure 2B). Among these antibodies, CAV-CF22 and CAV-CH76 exhibited strong binding affinities to both lineages against IBV-HA (Figure 2B). To preliminarily investigate the potential neutralizing capacity of these antibodies, we performed an HI assay (Figure 2C). Most of our isolated antibodies, except CAV-CG14, displayed HI activity for the tested IBVs, with both CAV-CF22 and CAV-CH76 exhibiting the broadest activity.

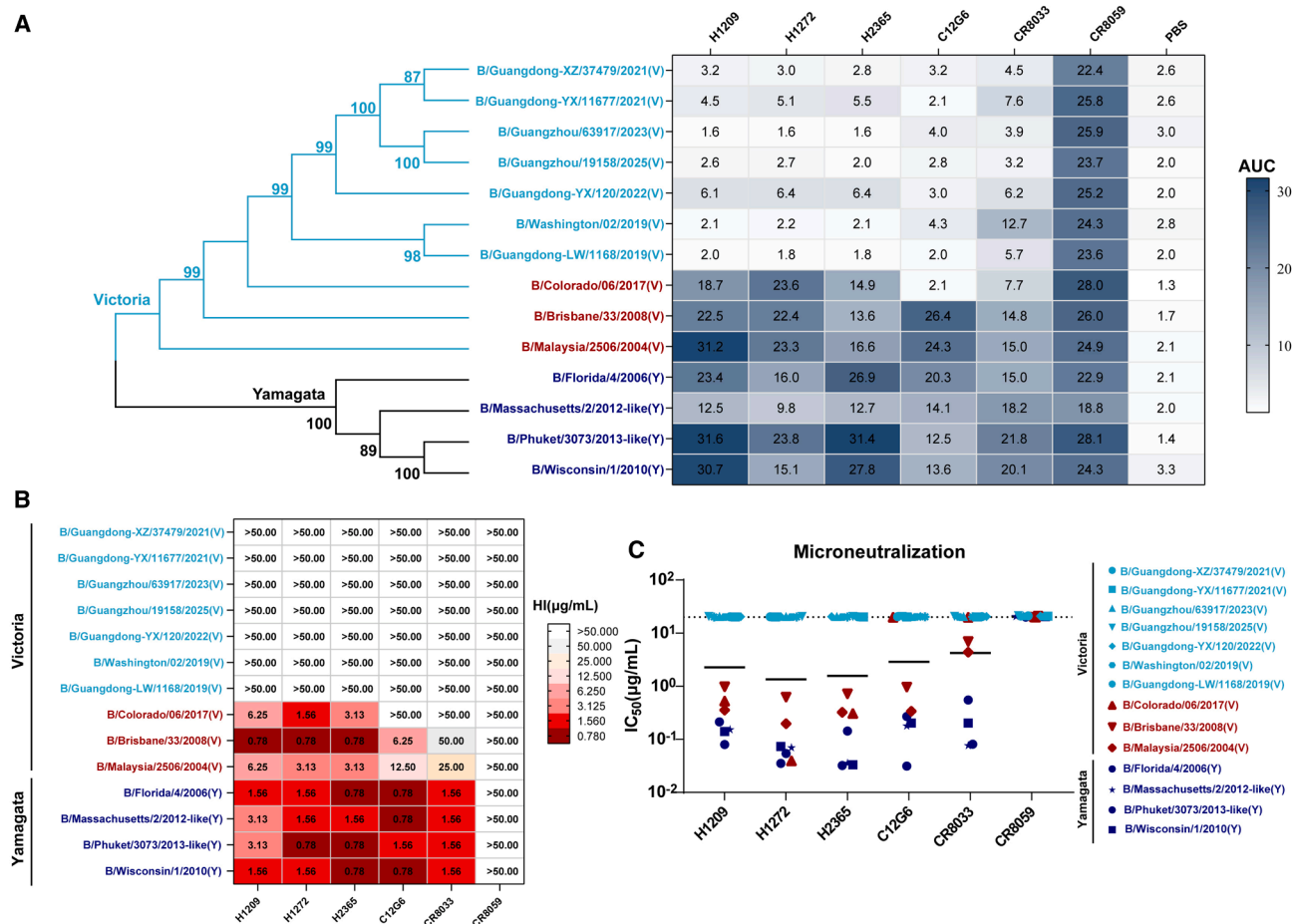


Figure 1. Previously reported broadly neutralizing HA head-directed mAbs fail to neutralize post-2019 IBV strains

(A) The binding ability of the reported antibodies obtained through enzyme-linked immunosorbent assay (ELISA) to IBV of two lineages. The binding of antibodies to IBV is reflected by the area under the curve (AUC). CR8059 is a positive control antibody. The 14 IBV strains used in this experiment were used to construct a phylogenetic tree using MEGA12.

(B) The neutralizing potential of antibodies targeting the HA head was reflected by the hemagglutination inhibition (HI) and was described by a heatmap. The darker the color, the lower the concentration of the antibody required to exert the HI effect.

(C) The IC_{50} values of the shown antibodies were determined by neutralization experiments using 14 IBV strains, with each virus strain plotted as a single symbol. IC_{50} values above 20 $\mu\text{g/mL}$ (dashed line) were scored as negative. The horizontal line denotes the mean.

Based on the above experimental results, we further validated the broad-spectrum properties and neutralizing capacity of the two antibodies, CAV-CF22 and CAV-CH76. CAV-CF22 exhibited strong binding ability to 13 out of the 14 selected strains, but it had relatively weak binding to B/Guangdong-XZ/37479/2021, whereas CAV-CH76 bound strongly to all 14 strains (Figure 2D). Although CAV-CF22 had weaker binding ability to the 4 selected Yamagata-lineage strains than CAV-CH76, it neutralized those 4 Yamagata strains more potently. Both antibodies displayed broad binding to diverse IBV strains, including post-2019 Victoria-lineage viruses—a reactivity profile absent in previously reported antibodies such as H1209, H1272, H2365, C12G6, and CR8033 (Figure 1A). Phylogenetic analyses showed that CAV-CF22 and CAV-CH76 retain neutralizing activity against viruses of the currently predominant V1A.3a.2 clade (Figures 2D and S1). Phylogenetic analysis via international ImMunoGeneTics information system (IMGT) shows CAV-CH76 comes from IGHV1-69/IGKV1D-12 and CAV-CF22 from

IGHV4-31/IGLV3-1 (Figure 2E; Table S3), unlike other broad-neutralizing antibodies against the IBV-HA RBS. Comparison with inferred germline sequences showed that both antibodies display relatively low variable heavy chain (VH) germline identity (CAV-CF22: 88.66% and CAV-CH76: 89.24%; Figure 2E), indicating extensive somatic hypermutation (SHM) and multiple rounds of affinity maturation. Biolayer interferometry (BLI) measurements confirmed strong binding affinities of both antibodies for representative HA proteins from both IBV lineages (Figure S2). In summary, the vaccine-induced antibodies, CAV-CF22 and CAV-CH76, exhibited cross-lineage binding and broad-neutralizing activity spanning Victoria- and Yamagata-lineage viruses *in vitro*.

Crystal structures of neutralizing antibodies bound to HA heads of IBV

To gain insights into the neutralizing mechanism of HA antibodies, we resolved the crystal structures of CAV-CH76 and

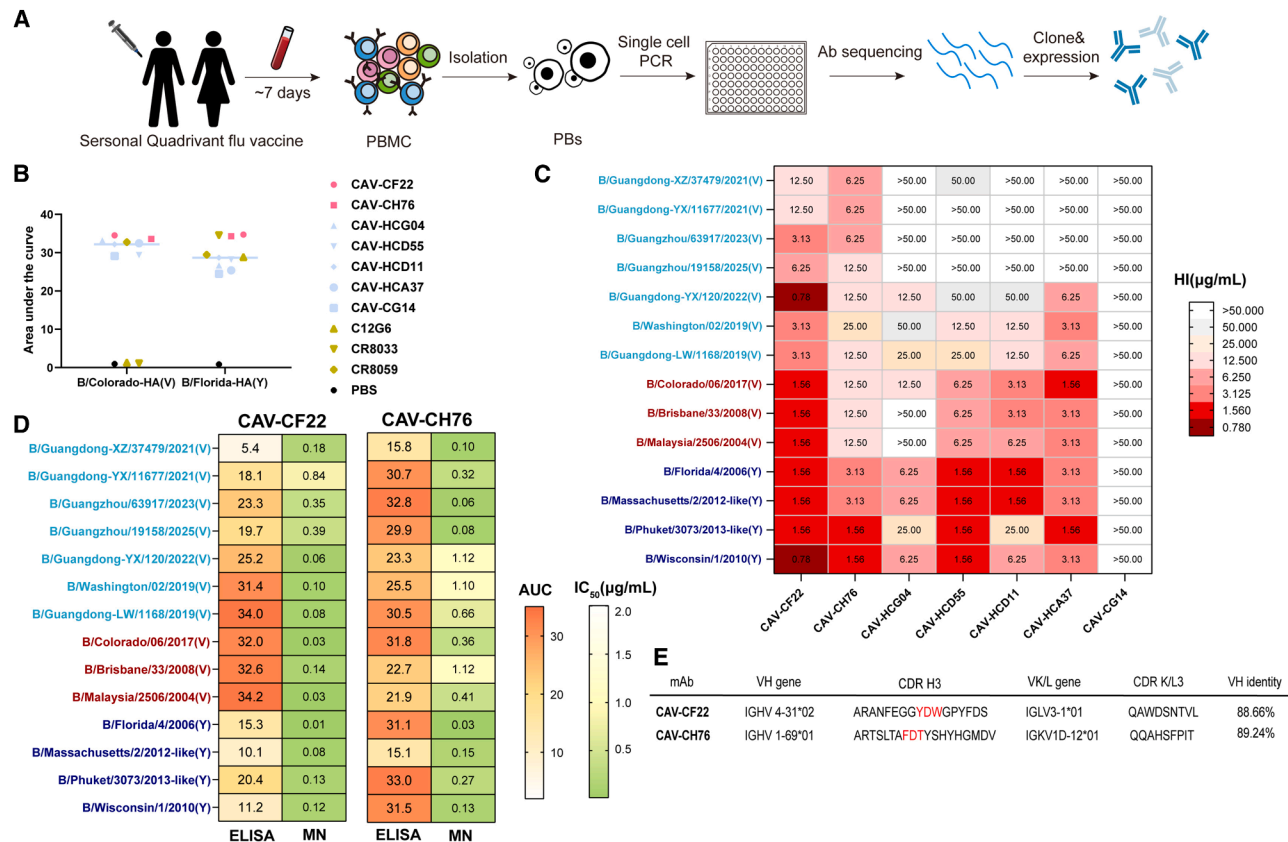


Figure 2. Influenza vaccination can induce broadly neutralizing antibodies in humans

(A) Flowchart of antibody screening.
 (B) The binding of antibodies to IBV-HA is reflected by the AUC. The larger the value, the stronger the antibody's binding ability to the recombinant protein. CR8059 is a positive control antibody. The horizontal line represents the median.
 (C) HI was described by a heatmap. The darker the color, the lower the concentration of the antibody required to exert the HI effect.
 (D) The binding and neutralization results of antibodies with 14 strains of virus are reflected through a heatmap.
 (E) Usage of immunoglobulin variable heavy- and light-chain genes and the HCDR3 sequences of HA mAbs.

CAV-CF22 Fabs bound to HA heads of IBV. The complex of CAV-CH76 bound to the HA head of B/Colorado/06/2017 was determined at 1.94 Å ($R_{\text{work}} = 0.234$ and $R_{\text{free}} = 0.283$), while that of CAV-CF22 bound to the B/Florida/4/2006 HA head was determined at 3.45 Å ($R_{\text{work}} = 0.262$ and $R_{\text{free}} = 0.297$) (Figures 3A, S3B, and S3C; Table S1). Due to the poor electron density of Fab constant regions, we only modeled the variable region of CAV-CF22 Fab. Both antibodies target the RBS region of the HA head (140 loop, 160 loop, 190 helix, and 240 loop) (Figures 3B, 3C, and S3A), which is similar to previously reported IBV RBS-targeting antibodies, like H1272.¹¹

Similar to the previously characterized antibody H1272,¹¹ CAV-CH76 predominantly utilizes its heavy chain to interact with HA, with only a few interactions from its light chain (Figures 3A and 3C). In detail, CAV-CH76 Fab utilizes its three complementarity-determining regions (CDRs) of heavy chain (HCDRs) to bind the RBS of HA, covering 93.7% of the epitope surface area (739.1 Å² out of 788.9 Å²), while only one amino acid (W32) on the light chain participates in the interaction (Figures 3C, 3D, and S3B). Among these interactions, HCDR3 has the largest interaction with HA, including regions in the 140

and 240 loops of RBS, as well as scattered epitopes in the β3 and β4 strands and the 190-helix (Figures 3C and 3D; Table S2). HCDR2 forms a salt bridge and additional contacts to the 160-loop and 190-helix (Figures 3C and 3D; Table S2). HCDR1 makes a limited number of contacts around the RBS, including interactions with β3 and the 140-loop (Figures 3C and 3D; Table S2). The W32 residue of light chain complementarity-determining regions 1 (LCDR1) interacts with the P239 residue of HA on the 240-loop. Among these interactions, six of ten hydrogen bonds were identified between HCDR3 and HA, three in HCDR1, and one in HCDR2 (Figure 3D; Table S2).

In contrast to CAV-CH76 and H1272, CAV-CF22 employs both its heavy and light chains to interact with the HA head (Figures 3C, 3E, and S3C), with a buried surface area of 512.2 Å² for heavy chains and 282.0 Å² for light chains. HCDR3 establishes substantial interactions with the 140-loop, 190-helix, and 240-loop of RBS and also forms some interactions out of RBS (F95 and D195) (Figures 3C and 3E; Table S2). HCDR2 predominantly interacts with the 140-loop and 240-loop, while only one residue, T32, on HCDR1 forms an interaction with P240 on the 240-loop of HA (Figures 3C

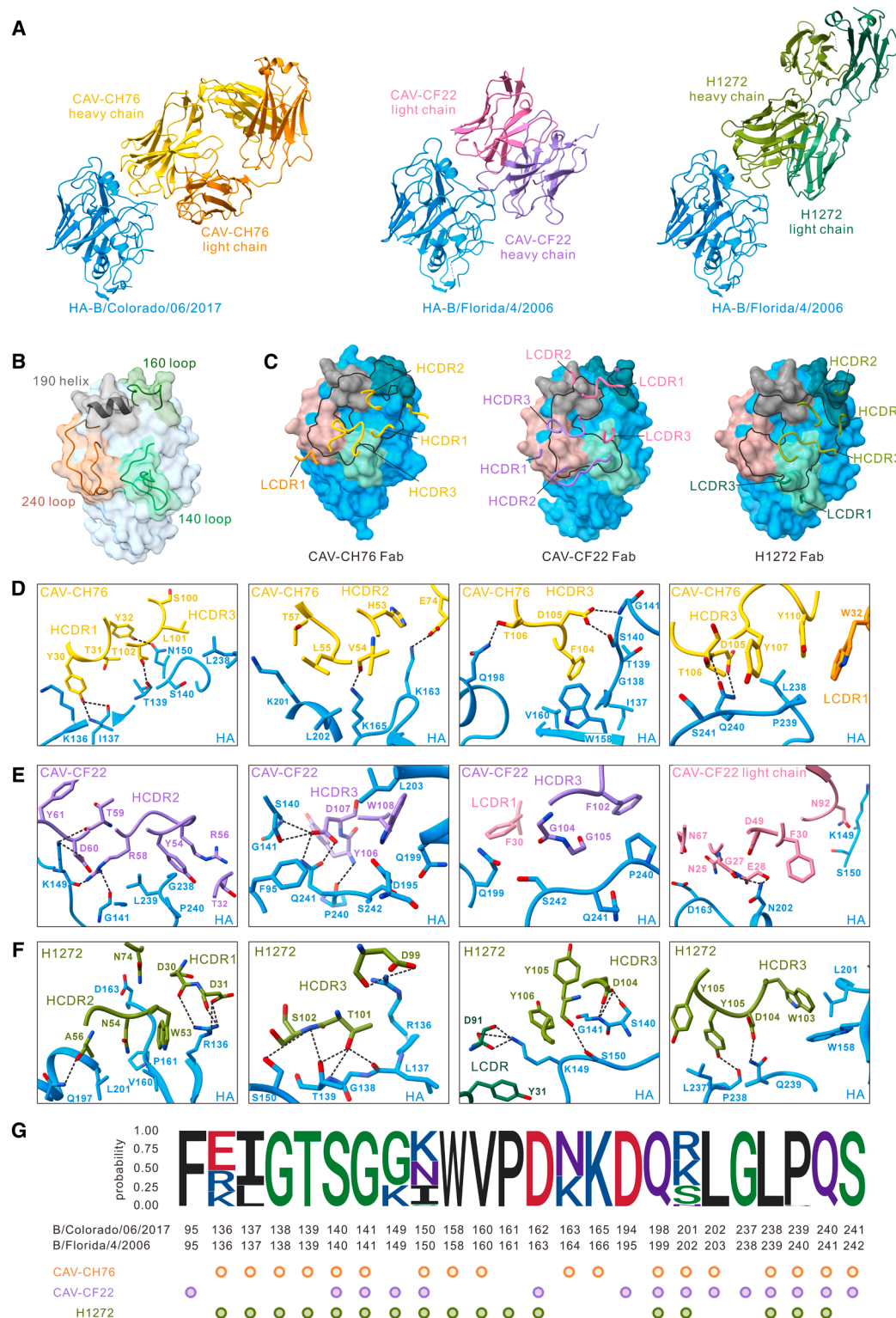


Figure 3. Complex Structures of antibodies bound to HA heads of pre-2019 IBVs

(A) The overall structures of antibodies complexed with HA heads of pre-2019 IBV (strains B/Colorado/06/2017 and B/Florida/4/2006) are presented. The HA heads are depicted in deep sky blue, while the Fabs CAV-CH76, CAV-CF22, and H1272 (PDB: 7KQG) are shown in yellow, purple, and green, respectively. The light chains of CAV-CH76 and H1272 are represented in darker shades of their respective colors, with the heavy chain of CAV-CF22 represented in darker shades of its respective color.

(B) The RBS region, including the 140-loop (spring green), 160-loop (forest green), 190-helix (gray), and 240-loop (brown), is shown on the HA head.

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and 3E; Table S2). LCDR1 mediates contacts with the 160-loop and 190-helix (Figures 3C and 3E; Table S2). Additionally, residues from LCDR2 and LCDR3 engage with K149, S150, and D163 of HA. Ten hydrogen bonds interactions were formed between CAV-CF22 Fab and HA head, with eight derived from the heavy chain (6 for HCDR3 and 2 for HCDR2) and two from LCDR1. One salt bridge was identified, involving the HCDR2 (Figure 3E; Table S2).

To assess epitope conservation, we analyzed approximately 70,000 sequences of IBV HA (1940–2024) from the Global Initiative for Sharing Avian Influenza Data (GISAID). The conservation analysis revealed that most of the binding epitopes utilized by CAV-CH76 and CAV-CF22 are highly conserved, with 72% and 80% of these epitopes displaying more than 90% conservation for CAV-CH76 and CAV-CF22, respectively (Figure 3G). Furthermore, over 70% of these epitopes exhibited more than 90% conservation in H1272 (Figure 3G). The results above imply that the binding mode of CAV-CH76 closely resembles that of H1272, primarily involving its heavy chain and displaying a similar epitope distribution on HA (Figures 3C, 3D, 3F, and 3G). By contrast, the binding mode of CAV-CF22 differs from that of H1272 (Figures 3C and 3E–3G).

Structural determinants of cross-neutralization against newly circulating Victoria-lineage strains with K/R136E mutations

CAV-CH76 and CAV-CF22 demonstrated broad-neutralizing activity against post-2019 IBV strains, whereas previously characterized antibodies (H1272, H2365, and H1209) showed diminished efficacy against recent isolates (Figures 1C and 2D). To investigate the structural basis for these differences, we aligned HA sequences (GISAID) from pre-2019 and post-2019 (2019–2025) Victoria-lineage and Yamagata-lineage (1988–2020) IBV strains (Figure 4A). Conservation analysis of key interaction residues identified three mutations: K/R136E, N150K, and a deletion at D162. Notably, a significant charge reversal mutation occurred at residue 136, where K (pre-2019 Victoria) or R (Yamagata) was replaced by E in post-2019 Victoria-lineage strains (Figure 4A).

Based on the solved complex structures, we further performed detailed interaction analysis toward these three mutated sites. Both previously reported antibodies and our antibodies displayed similar binding patterns with the N/S/I 150 or D162/163 of HA, and each antibody exhibited only a limited number of interactions with them, suggesting a conserved and relatively weak engagement at these sites (Figure 4B). By contrast, at residue 136, the binding patterns differed significantly between previously reported neutralizing antibodies (H1209, H1272, and H2365) and our newly identified antibodies (CAV-CH76 and CAV-CF22). CAV-CH76 only formed a few contacts with residue 136, whereas CAV-CF22 exhibited no direct interactions within a

distance cutoff of 4.0 Å. However, H1272, H2365, and H1209 depended heavily on interactions mediated by R136, forming 37, 28, and 13 contacts, respectively, along with a network of hydrogen bonds and salt bridges (Figures 4B and 4C). We therefore propose that the K136E mutation in post-2019 HA abolished the binding and neutralization of previously reported antibodies while having no impact on the activity of our antibodies.

To confirm our hypothesis, we first performed an ELISA assay and found that both CAV-CF22 and CAV-CH76 could bind HA heads of post-2019 (B/Guangdong-YX/120/2022) with high affinity. By contrast, there are no detectable interactions between previously reported antibodies and post-2019 HA heads (Figure S4A). Furthermore, we determined the complex structures of CAV-CH76 and CAV-CF22 Fab in complex with post-2019 HA (B/Guangdong-YX/120/2022) at a resolution of 3.01 and 2.98 Å, respectively (Figures 4D and S4C–S4F; Table S1). The overall binding patterns of these two antibodies on post-2019 HA are nearly identical to that of pre-2019 (Figure S6). Further analysis of interaction on E136 revealed it formed ~12 (out of a total of 106) contacts with CAV-CH76 but none with CAV-CF22 (Figures 4B and 4E). Meanwhile, no hydrogen bond or salt bridge interaction was observed between E136 and either antibody (Figures 4B and 4E). Similar to the CAV-CF22-pre-2019 HA complex, CAV-CF22 showed no obvious interaction with E136 in the post-2019. By contrast, CAV-CH76 exhibited slightly increased interactions with E136 in post-2019 compared with K136 in pre-2019 HA (rising from 7 to 12 contacts). All these data suggested that K136E has no effect on the binding of our antibodies to post-2019 HA.

To investigate why previously reported mAbs fail to neutralize recent strains, we validated key residue hotspots. We introduced two interface mutations, K136E and N150K, into the HA head of B/Colorado/06/2017. ELISA data suggested that the K136E mutation alone was sufficient to completely abolish binding for all previously reported antibodies, whereas the N150K mutation had no such effect (Figure S4B). Other mutations, such as D129, G133, and K203, are far away from the binding interface and are therefore unlikely to impact antibody binding. In summary, the K136E mutation is the primary determinant for the loss of neutralization against post-2019 influenza strains.

Aspartic acid-mediated sialic acid mimicry underlies neutralization

Similar to H1272, CAV-CH76 and CAV-CF22 feature a conserved aspartic acid residue at the HCDR3 tip that deeply penetrates the RBS pocket to establish multiple interactions with HA.¹¹ Structural analysis demonstrates that the aspartic acids (D105 in CAV-CH76, D107 in CAV-CF22, and D104 in H1272) at the HCDR3 tip form hydrogen bonds with HA residues S140 and/or G141, mimicking the carboxylate group of sialic

(C) The epitopes of CAV-CH76 Fab, CAV-CF22 Fab, and H1272 Fab are imprinted on HA heads. The HA heads are shown on the surface, on which the epitope of antibodies is footprinted with a black line. The colors of the heavy and light chains correspond to those used in (A).

(D–F) Detailed interactions between the CAV-CH76 Fab (D), CAV-CF22 Fab (E), and H1272 Fab (F) and the HA heads are illustrated, with hydrogen bonds and salt bridges represented by dashed lines.

(G) The conservation analysis of interaction epitopes on IBV HA. The residues that interact with the antibodies are marked by colored circles. The x axis represents the interaction epitopes on the HA head, while the y axis indicates the level of amino acid conservation. The size of each letter representing an amino acid signifies the degree of conservation, with larger letters indicating higher conservation levels.

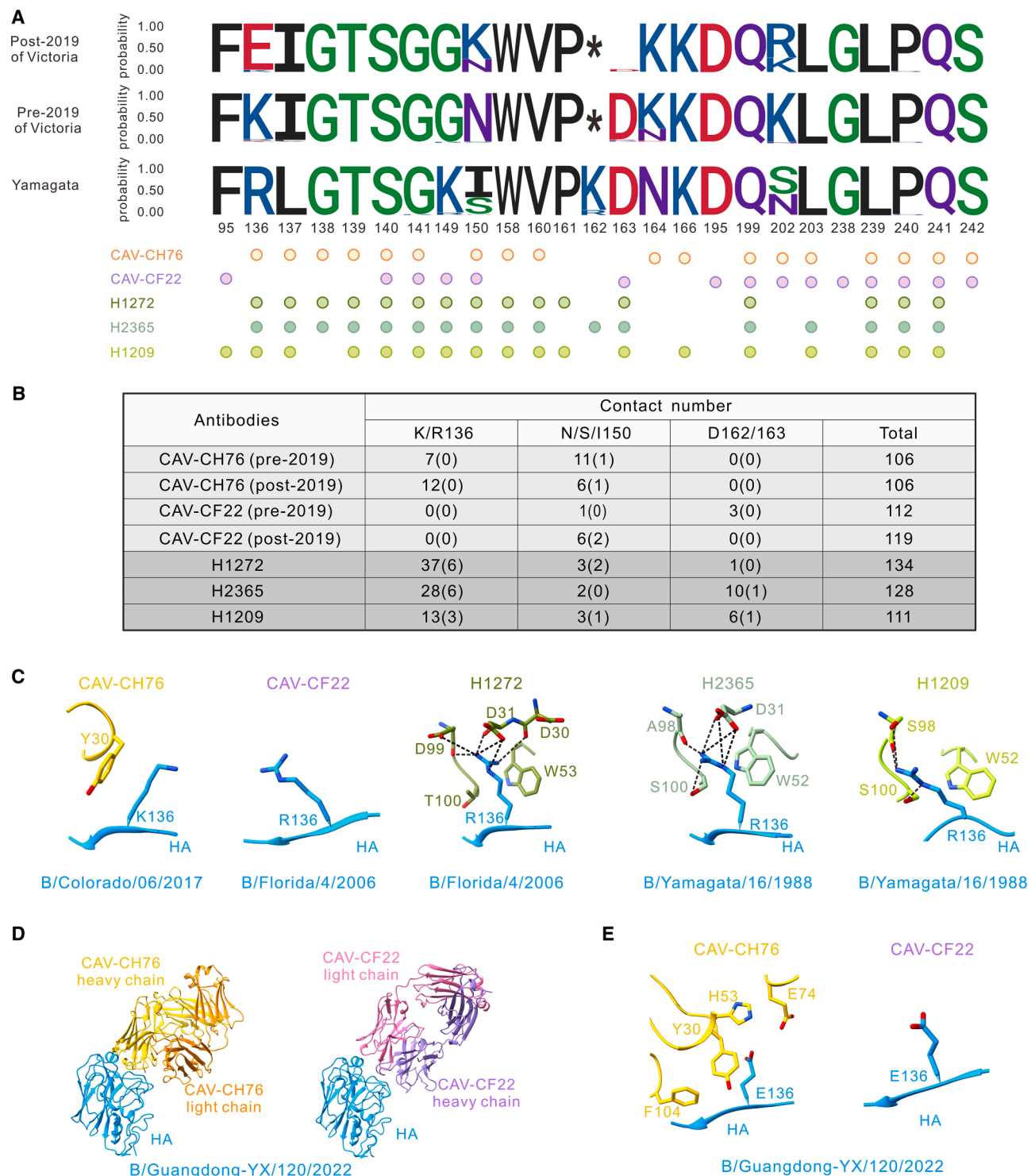


Figure 4. The structural basis of antibodies enables to cross-neutralize currently circulating IBVs

(A) The epitope conservation analysis of antibodies on the HA head (pre- and post-2019 of Victoria lineage and Yamagata lineage). The residues that interact with the antibodies are marked by colored circles. The symbol * denotes the absence of a corresponding amino acid at that site.

(B) The number of interactions between HA K/R136, N/S/I 150, D162/163, and antibodies is summarized. A distance cutoff of 4 Å was used, with the values in parentheses representing the number of hydrogen bonds and salt bridges.

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acid (Figures 5A–5F; Table S2). The residues before and after the central Asp—¹⁰⁴FD¹⁰⁶ (CAV-CH76), ¹⁰⁶YDW¹⁰⁸ (CAV-CF22), and ¹⁰³WDY¹⁰⁵ (H1272)—further engage the RBS interaction but display distinct interaction modes from that of sialic acid's acetamido and hydroxyl groups. Comparative analysis indicates that while the neutralizing antibodies of IBV employ similar strategies to IAV, distinct differences exist for the sialic acid mimicry. For IAV, the critical Asp and the residue before Asp were found to precisely mimic carboxylate and acetamido groups of sialic acid, respectively.^{21,22}

We further conducted a neutralization assay to confirm the critical role of HCDR3 Asp (Figures 5G–5I). Consistent with structural data, CAV-CH76-D105A and CAV-CF22-D107A completely lost their neutralizing activity against B/Colorado/06/2017 and B/Florida/4/2006, respectively (Figures 5G and 5H). Interestingly, the neutralizing activity of CAV-CF22-D107A still remains at a relatively potent level toward B/Colorado/06/2017, yet it decreases heavily compared with wild-type CAV-CF22 (Figure 5G). By contrast, CAV-CH76-D105A heavily diminished its neutralizing activity against B/Florida/4/2006 (Figure 5H). It is notable that both of them have lost their neutralizing activity against B/Guangdong-YX/120/2022, the IBV strain post-2019 (Figure 5I). These findings demonstrate that while both antibodies mimic sialic acid via a conserved HCDR3 Asp, their binding modes diverge in residue-specific interactions. Particularly, the aspartic acid residue at the tip of HCDR3 plays a pivotal role in HA binding, with its surrounding residues together contributing to the overall binding and neutralizing capacity.

Furthermore, we conducted structure-guided single-site alanine scanning mutations on HA and performed an ELISA binding assay to identify the key residues involved in antibody binding. For B/Florida/4/2006 HA, substitutions of F95A, S140A, D195A, Q199A, L203A, L239A, P240A, Q241A, and S242A markedly reduced binding with CAV-CF22. For B/Colorado/06/2017 HA, substitutions of S140A, G141A, W158A, V160A, L202A, L238A, and Q240A substantially reduced binding with CAV-CH76, whereas other tested substitutions had only minor effects (Figure S7). Retrospective analysis of our solved complex structures revealed that most of these key HA residues mainly interact with HCDR3, which reflected the core neutralizing mechanism of these antibodies.

CAV-CH76 and CAV-CF22 confer prophylactic and therapeutic protection against lethal seasonal influenza B challenge *in vivo*

To evaluate the prophylactic and therapeutic antiviral activities of different antibody doses *in vivo*, this study employed three mouse-adapted IBV strains (MA-B/Florida/4/2006(Y), MA-B/Colorado/06/2017(V), and MA-B/Guangdong-YX/120/2022(V)) for subsequent investigations. Both antibodies demonstrated robust prophylactic and therapeutic efficacy against lethal challenge by all three IBVs (Figures 6 and S8). In the prophylactic groups, CAV-CF22 at ≥ 0.5 mg/kg completely protected mice from infection with each of the three viruses, and body weight

in these groups remained essentially at baseline levels (Figures 6C, 6G, 6K, and S8). CAV-CH76 at ≥ 0.5 mg/kg conferred complete protection against lethal challenge by the Yamagata-lineage and the post-2019 Victoria-lineage IBVs. The lower dose of CAV-CH76 (0.5 mg/kg) provided 80% prophylactic protection against lethal challenge by the B/Colorado/06/2017 (Figures 6D, 6H, and 6L). In the therapeutic groups, mice treated with CAV-CF22 or CAV-CH76 at ≥ 2 mg/kg 1 day after infection were fully protected from lethal challenge by all three IBVs (Figure 6). Lower therapeutic doses (as low as 0.5 mg/kg) still afforded partial protection for each virus (Figure 6).

To benchmark efficacy, we directly compared CAV-CF22 and CAV-CH76 (2 mg/kg) to two well-characterized bnAbs, CR9114 and CR8033. At 2 mg/kg, CAV-CF22 and CAV-CH76 conferred complete protection across the three viruses tested. By contrast, CR9114 and CR8033 showed variable protection depending on the challenge strain: CR8033 protected against B/Florida/4/2006 but not B/Colorado/06/2017, whereas CR9114 protected poorly against B/Florida/4/2006 (only $\sim 20\%$ survival) but protected against the B/Guangdong-YX/120/2022 strain (Figures 6E, 6I, and 6M). H1272 was included as an IBV mAb control. Consistent with the *in vitro* functional data, H1272 fully protected mice from lethal challenge with pre-2019 dual-lineage viruses but failed to confer protection against post-2019 Victoria-lineage virus infection (Figures 6 and S8).

Coinciding with the high survival rates, both prophylactic and therapeutic antibody administration led to a significant reduction in lung viral load compared with the PBS control (Figures S9A and S9B). Furthermore, histopathological analysis showed that prophylactic treatment with CAV-CF22 and CAV-CH76 markedly attenuated IBV-induced lung injury and associated inflammation (Figure S9C). While the potent neutralization activity of the antibodies is presumed to be the primary driver of protection, Fc-effector function may also contribute to the overall therapeutic mechanism. Specifically, both CAV-CF22 and CAV-CH76 elicited concentration-dependent antibody-dependent cellular cytotoxicity (ADCC) activity against infected cells (Figures S9D–S9F).

DISCUSSION

IBV remains a significant threat to global public health, causing considerable disease and economic burdens worldwide every year. Here, we have identified two human monoclonal antibodies (CAV-CH76 and CAV-CF22) from vaccinated individuals, which can effectively neutralize a variety of IBV strains from the Victoria and Yamagata lineages isolated over the past two decades. Both antibodies demonstrated effective prophylactic and therapeutic effects in lethal murine challenge models. Specifically, CAV-CF22 and CAV-CH76 have a preventive effect against B/Victoria-lineage viruses with position 136 mutations that emerged after 2019. These findings indicate that influenza vaccination can induce the production of protective antibodies in the body that target IBV that have undergone antigenic drift.

(C) Detailed interactions between residue 136 of HA and antibodies are illustrated, with hydrogen bonds and salt bridges represented by black dashed lines.

(D) The overall structures of antibodies CAV-CH76/CAV-CF22 Fab complexed with the HA head of IBV (strains B/Guangdong-YX/120/2022) are presented.

(E) Detailed interactions between residue E136 of HA (B/Guangdong-YX/120/2022) and antibodies (CAV-CH76 and CAV-CF22) are illustrated, with hydrogen bonds and salt bridges represented by black dashed lines.

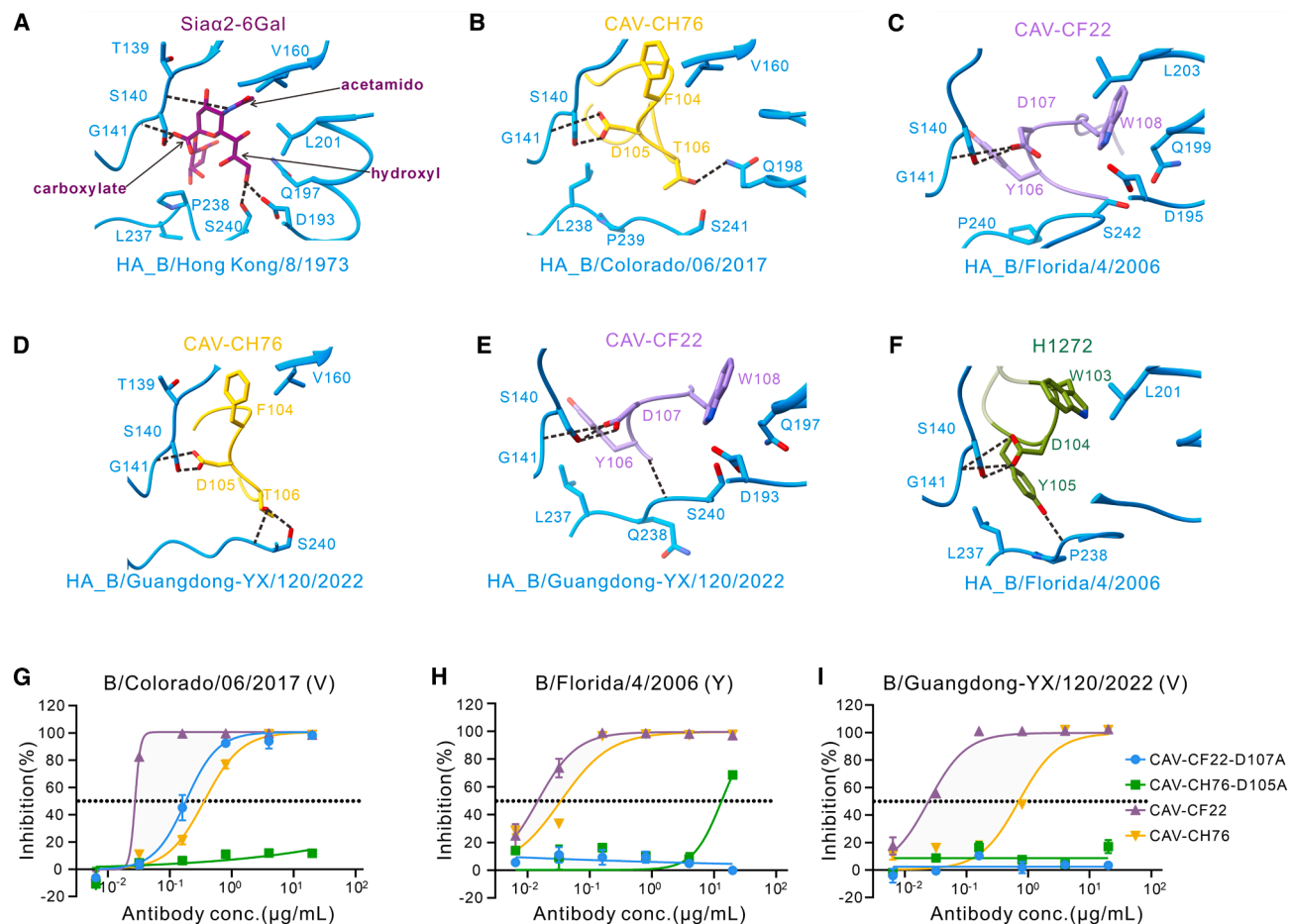


Figure 5. The antibodies bind to IBV HA by sialic acid mimicry

(A–F) The interaction details of sialic acid (PDB: 2RFU) (A), CAV-CH76 (B and D), CAV-CF22 (C and E), and H1272 (PDB: 7KQG) (F), bound to HA heads are illustrated. The critical interacting residues are depicted using sticks. The hydrogen bonds are shown by dashed lines.

(G–I) Neutralization ability of single-point mutated antibodies against IBVs.

We found that the broadly neutralizing antibodies previously isolated against the IBV-HA RBS region showed significantly reduced or even no binding and neutralization against the B/Victoria lineage after 2019. The global absence of B/Yamagata-lineage detection since March 2020 underscores the urgent need for Victoria-inclusive bnAbs.^{16–18} Existing candidates, such as C12G6 and CR8033, show limited breadth and lack epitope-resolution data, particularly against post-2019 Victoria-lineage viruses.^{12,13} While H1209, H1272, and H2365 target the RBS of IBV HA, their functional validation *in vitro* and *in vivo* remains incomplete.¹¹ By contrast, CAV-CF22 and CAV-CH76 surpass these benchmarks by neutralizing all tested IBV strains (2004–2025), including the currently predominant V1A.3a.2 clade.

As antibodies undergo affinity maturation, their levels of SHM increase, which can enhance binding affinity and broaden neutralization breadth,²³ as demonstrated for HB420 and 3114, whose SHM enabled cross-reactivity beyond their germline specificities.^{24,25} In the study, the high SHM levels observed in CAV-CF22 and CAV-CH76 suggest that they have extensive SHM and multiple rounds of affinity maturation. Given the marked antigenic drift of the B/Victoria lineage after 2019 and

the reduced breadth of some previously reported RBS-targeting antibodies, it is plausible that the updated B/Washington/02/2019 vaccine boosted these memory lineages, promoting further affinity maturation and enabling the broad cross-lineage neutralization reported here.

The previously characterized antibodies had weak or no binding and neutralization effects on post-2019 IBV strains. According to the World Influenza Centre's report (<https://www.crick.ac.uk/research/platforms-and-facilities/worldwide-influenza-centre/annual-and-interim-reports>), the B/Victoria-lineage virus underwent a mutation at several sites during 2018 and 2019, including K136E.²⁶ K136 is localized within the interface between antibody and HA. While the K136E substitution abrogated the binding and neutralization by historical RBS-targeting antibodies (H1272, H2365, and H1209)—which depend on hydrogen bonds and salt bridges with K136—CAV-CF22 and CAV-CH76 still retained potency against K136E variants. Structural analysis clarified this divergence: CAV-CF22 avoids direct interaction with residue 136, while CAV-CH76 engages it minimally (Figure 4). The K136E mutation had no measurable impact on the binding affinity of CAV-CF22 and only a minimal effect on CAV-CH76.

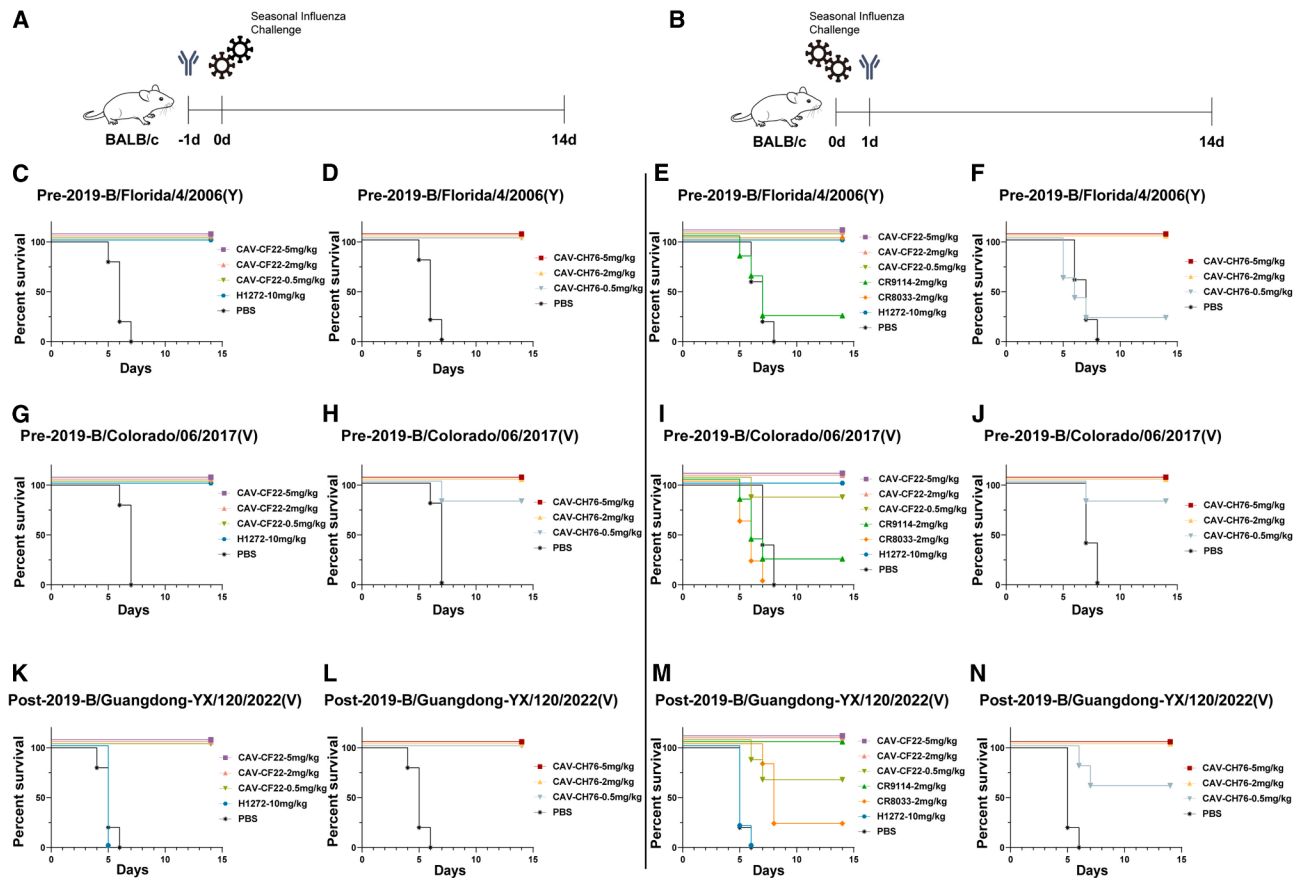


Figure 6. CAV-CH76 and CAV-CF22 confer prophylactic and therapeutic protection against lethal seasonal influenza B challenge *in vivo*
(A and B) Pre-exposure prophylaxis and post-exposure therapeutic studies. BALB/c mice received the indicated doses of mAbs intraperitoneally 24 h before or 24 h after virus challenge. The challenge dose was 10 lethal dose 50 (LD_{50}) of virus, and the experiment ended on day 14 post-challenge.
(C–F) Challenge with B/Florida/4/2006 (Yamagata). Pre-exposure prophylaxis with graded doses of CAV-CF22 and 10 mg/kg H1272 (C) and CAV-CH76 (D). Post-exposure therapy with CAV-CF22 and 2 mg/kg CR9114 and CR8033 (E) and CAV-CH76 (F).
(G–J) Challenge with B/Colorado/06/2017 (Victoria). Pre-exposure prophylaxis with graded doses of CAV-CF22 and 10 mg/kg H1272 (G) and CAV-CH76 (H). Post-exposure therapy with CAV-CF22 and 2 mg/kg CR9114 and CR8033 (I) and CAV-CH76 (J).
(K–N) Challenge with B/Guangdong-YX/120/2022 (Victoria). Pre-exposure prophylaxis with graded doses of CAV-CF22 and 10 mg/kg H1272 (K) and CAV-CH76 (L). Post-exposure therapy with CAV-CF22 and 2 mg/kg CR9114 and CR8033 (M) and CAV-CH76 (N).

CAV-CF22 and CAV-CH76 were generated using a vaccine containing the HA component of a Victoria lineage (B/Washington/02/2019-like virus), which has a K to E mutation at position 136. Both antibodies target the HA RBS, and similar to CR8033 and C12G6, aspartic acid (D105/D107) on HCDR3 of CAV-CH76 and CAV-CF22 mimics sialic acid via hydrogen bonding (Figure 5). Their superior HI breadth suggests divergent epitope recognition. CAV-CH76 and CAV-CF22 engage conserved RBS residues through distinct paratope architectures (72% of epitope residues for CAV-CH76 and 80% for CAV-CF22 are >90% conserved across lineages) (Figure 3). To further understand the key residues recognized by antibodies, we selected antibody escape mutants by passaging IBV in the presence of CAV-CF22 or CAV-CH76. Under CAV-CF22 pressure, B/Florida/4/2006 HA acquired a G141R substitution, whereas under CAV-CH76 pressure, B/Colorado/06/2017 HA acquired a G141E substitution (Figure S10), consistent with crystal structures showing G141 within the core epitope of both antibodies.

To date, the structural data for antibodies targeting the HA RBS of IBV remain limited, encompassing only the previously reported antibodies (H1272, H2365, and H1209) and our newly isolated antibodies (CAV-CH76 and CAV-CF22). While all these antibodies target the RBS region and share similar binding epitopes, key distinctions exist. The H1272, H2365, H1209, and CAV-CH76 interactions with HA are mediated primarily by heavy-chain contacts. By contrast, the light chain of CAV-CF22 also contributes to HA engagement and forms part of the antibody-HA interface (Figure S5). This is reflected in their epitopes: CAV-CF22 binds preferentially to the C-terminal region of the HA head, whereas the others map to the N-terminal region. Furthermore, detailed interaction analysis revealed that in complexes with previously reported antibodies, particularly H1272 and H2365, residue 136 serves as a major interaction hotspot. This is significant, as the K136E substitution in post-2019 IBV strains confers immune escape from these antibodies. By contrast, the interaction patterns of CAV-CF22 and CAV-CH76 do not rely on any single

residue. We hypothesized that the more balanced binding mode may therefore be less susceptible to viral escape.

Although we cannot determine the situation of the generation of broadly protective antibodies after influenza vaccination at the population level, the volunteers in the experiment can induce protective antibodies against mutant strains through influenza vaccination, so we recommend prevention through influenza vaccination. At the same time, the structural and functional insights gained from CAV-CH76 and CAV-CF22 offer dual translational promise. Firstly, their structural definition epitopes provide a blueprint for engineering IBV vaccines that focus immunity on conserved RBS vulnerabilities. Secondly, the antibodies' pan-lineage efficacy supports their clinical development as pre- or post-exposure biologics, particularly for high-risk populations where vaccine mismatches or antiviral resistance arise. Future work should explore combinatorial regimens with stalk-targeting bnAbs to minimize escape mutant emergence.

A limitation of this study is the reliance on the murine model for *in vivo* efficacy assessment. Due to institutional and logistical constraints, we were unable to incorporate a ferret challenge model, which is recognized for its high translational relevance to human influenza pathogenesis.

RESOURCE AVAILABILITY

Lead contact

Requests for further information and resources should be directed to and will be fulfilled by the lead contact, Yao-Qing Chen (chenyaoqing@mail.sysu.edu.cn).

Materials availability

All virus strains and unique reagents used in this study can be obtained from the [lead contact](#), dependent upon availability.

Data and code availability

- The atomic structures of the antibody-HA head complexes have been deposited in the Protein Data Bank (PDB: 9JNO, 9L8Z, 9XTH, and 9XSX). The Next-Generation Sequencing (NGS) sequencing data have been deposited in NCBI Sequence Read Archive (SRA) (NCBI SRA: PRJNA1424181).
- This work did not use or generate new code.
- Any additional information required to reanalyze the data reported in this paper is available from the [lead contact](#) upon request.

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AUTHOR CONTRIBUTIONS

Y.-Q.C., J.G., and Y.-L.S. conceptualized, coordinated, and supervised the study. X.C., Y. Zhai, and M.F. designed and performed experiments and wrote the manuscript. J.G., M.F., and M. Zhao performed crystal structure determi-

nation and analysis. W.L., B.H., Z.H., D.W., M. Zheng, Z.Z., K. Li, M.X., L.L., K. Lv, and S.Z. assisted in the antibody screening and purification. S.L. assisted with sequence conservation analysis. Y.S., R.Z., and X.Z. assisted in the recombinant antigen construction and purification. Y.K., H. Lu, X.Y., Y. Zhang, L.Y., N.Z., H. Liang, Y.C., and F.Y. participated in the preliminary sample collection. All authors have read and approved the article.

DECLARATION OF INTERESTS

The authors declare no competing interests.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE WRITING PROCESS

The authors utilized ChatGPT 3.5 to enhance grammar and readability during the manuscript writing. Following this, the authors thoroughly reviewed and edited the article, assuming full responsibility for its content.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Pacific Blue™ anti-human CD19 Antibody	BioLegend	Cat# 302224; RRID: AB_493653
APC anti-human CD27 Antibody	BioLegend	Cat# 302810; RRID: AB_314302
PE anti-human CD38 Antibody	BioLegend	Cat# 303506; RRID: AB_314358
Peroxidase-AffiniPure Goat Anti-Human IgG, Fcγ Fragment Specific	Jackson ImmunoResearch Labs	Cat# 109-035-098; RRID: AB_2337586
IBV-NP Antibody	In house	N/A
Bacterial and virus strains		
B/Florida/4/2006 (Yamagata)	In house	N/A
B/Wisconsin/1/2010 (Yamagata)	In house	N/A
B/Massachusetts/2/2012-like virus (Yamagata)	In house	N/A
B/Phuket/3073/2013-like virus (Yamagata)	In house	N/A
B/Malaysia/2506/2004 (Victoria)	In house	N/A
B/Brisbane/60/2008 (Victoria)	In house	N/A
B/Colorado/06/2017 (Victoria)	In house	N/A
B/Washington/02/2019 (Victoria)	In house	N/A
B/Guangdong-Liwan/1168/2019 (Victoria)	In house	N/A
B/Guangdong-YX/11677/2021 (Victoria)	In house	N/A
B/Guangdong-XZ/37479/2021 (Victoria)	In house	N/A
B/Guangdong-YX/120/2022 (Victoria)	In house	N/A
B/Guangzhou/63917/2023 (Victoria)	In house	N/A
B/Guangzhou/19158/2025 (Victoria)	In house	N/A
Biological samples		
Human PBMC	This paper	N/A
Mouse lung tissue samples	This paper	N/A
Chemicals, peptides, and recombinant proteins		
B/Florida/4/2006 (Yamagata) HA	This paper	EPI159978
B/Colorado/06/2017 (Victoria) HA	This paper	EPI1056637
B/Guangdong-YX/120/2022 (Victoria) HA	In house	Table S5
Polyethylenimine, Linear, MW 25000, Transfection Grade (PEI 25K)	Polysciences	Cat# 23966-1
ABTS	Sigma-Aldrich	Cat# A1888
Trypsin from bovine pancreas	Sigma-Aldrich	Cat# T8802
rProtein A Beads 4FF	Smart-lifesciences	Cat# SA015100
Ni Sepharose High Performance	Cytiva	Cat#17526802
Dulbecco Modified Eagle's Medium (DMEM)	Gibco	Cat# C11995500BT
Fetal bovine serum (FBS)	Gibco	Cat# A5669701
RPMI Medium 1640 basic (1640)	Gibco	Cat# C11875500BT
Fetal Bovine Serum (Defined)	ABW	Cat# AB-FBS-IG-0050
Sf-900™ II SFM	Gibco	Cat# 10902088
Penicillin-Streptomycin Solution	Gibco	Cat# 15140-122
Blasticidin S, Sterile	(MCE)	Cat# HY-K1054-5mL
Zeocin™	ThermoFisher	Cat# R25001
Lymphoprep™	Serumwerk Bernburg	Cat# 1858
Cellfectin II Reagent	Invitrogen	Cat# 10362-100

(Continued on next page)

Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
10K-cutoff concentrator	Merck millipore	Cat# UFC801096
30K-cutoff concentrator	Merck millipore	Cat# UFC910096
Superdex 200 Increase columns 10/300GL	Cytiva	Cat# 28990944
Flat bottom 96-well plates	NUNC	Cat# 442404
Bovine Serum Albumin(BSA)	GBCBIO	Cat# 0332-100g
Tribromoethanol, TBE	Meilunbio	Cat# MA0478

Critical commercial assays

QUANTI-Luc™ 4 Lucia/Gaussia	InvivoGen	Cat# rep-qlc4lg1
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Deposited data

CAV-CH76 Fab/HA head (B/Colorado/06/2017) complex structure	Protein Data Bank (PDB) https://www.rcsb.org/	PDB: 9JNO
CAV-CF22 Fab/HA head (B/Florida/4/2006) complex structure	Protein Data Bank (PDB) https://www.rcsb.org/	PDB: 9L8Z
CAV-CH76 Fab/HA head (B/Guangdong-YX/120/2022) complex structure	Protein Data Bank (PDB) https://www.rcsb.org/	PDB: 9XTH
CAV-CF22 Fab/HA head (B/Guangdong-YX/120/2022) complex structure	Protein Data Bank (PDB) https://www.rcsb.org/	PDB: 9XSX
NGS raw data	NCBI Sequence Read Archive (SRA) https://www.ncbi.nlm.nih.gov/sra	NCBI SRA: PRJNA1424181

Experimental models: Cell lines

HEK293T	ATCC	N/A
MDCK cells	ATCC	N/A
Sf9	ATCC	N/A
Jurkat-Lucia™ NFAT-CD16 cells	InvivoGen	Cat# jktl-nfat-cd16

Experimental models: Organisms/strains

Mouse: Wild type (BALB/c mice)	BesTest	Cat# BALC8W
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Software and algorithms

GraphPad Prism (version 8.0)	GraphPad Software	https://www.graphpad.com/
Pymol (version 2.2.5)	PyMOL	https://www.pymol.org/
ChimeraX (version 1.7.1)	UCSF ChimeraX	https://www.cgl.ucsf.edu/chimerax/
Coot (version 0.9.2 EL)	Coot	https://www2.mrc-lmb.cam.ac.uk/personal/pemsley/coot/
Phenix (version 1.16-3549)	PHENIX	https://phenix-online.org
MEGA 12(version12.1)	MEGA	https://www.megasoftware.net/
R (version 4.1.0)	R Core team	N/A

Other

BD FACSAria™ III	BD	BD FACSAria™ III
MD SpectraMax ABS plus	MD	SpectraMax ABS plus
Octet R8 Protein Analysis System	Sartorius	Octet R8

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Human sample donors

This study protocol was reviewed and approved by the Medical Ethics Committee of the School of Public Health, Sun Yat-sen University (Ethics Code: School of Public Health of SYSU Medical Ethics [2019] No. 039). The informed consent was obtained from the participant. Detailed information of volunteers is provided in [Table S4](#).

Mouse models

All animal experiments were conducted in accordance with the protocol approved by the Institutional Animal Care and Use Committee (IACUC) of Sun Yat-sen University (Approval No: SYSU-IACUC-2019-B095). 6–8-week-old healthy wild-type female BALB/c mice were purchased from BesTest for this experiment. Randomly assign all enrolled mice to each group, with five mice per group. Five

mice were housed in standard cages. All mice were housed in pathogen-free rooms under 12-hour light/dark cycles with clean bedding and free access to food and water. Cages and bedding were changed weekly. Mice received humane care and were handled using sterile techniques in a biosafety cabinet.

Cell lines

MDCK, HEK293T cells were cultured at 37 °C with 5% CO₂ in Dulbecco's Modified Eagle medium (DMEM, Gibco), supplemented with 10% fetal bovine serum (FBS, Gibco) and 1% Penicillin-Streptomycin Solution (Gibco). Sf9 (*Spodoptera frugiperda* 9) cells were maintained in Sf-900™ II SFM medium (Gibco) at 27 °C. Hi 5 cells were maintained in SIM HF expression medium (Sino Biological) at 27 °C. Jurkat-Lucia™ NFAT-CD16 cells were cultured in RPMI Medium 1640 basic (1640, Gibco) supplemented with 10% fetal bovine serum (FBS, Gibco), 1% penicillin-streptomycin solution (Gibco), 10 µg/mL Blasticidin S, Sterile (MCE), and 100 µg/mL Zeocin™ (ThermoFisher) at 37 °C with 5% CO₂. Cells were routinely tested for mycoplasma contamination using culture-based methods.

Virus isolates

Wild type influenza virus B/Florida/4/2006, B/Malaysia/2506/2004, B/Washington/02/2019, B/Phuket/3073/2013-like virus, B/Wisconsin/1/2010, B/Brisbane/60/2008, B/Massachusetts/2/2012-like virus, B/Colorado/06/2017, B/Guangdong-YX/11677/2021, B/Guangdong-Liwan/1168/2019, B/Guangdong-YX/120/2022, B/Guangzhou/63917/2023, B/Guangzhou/19158/2025 and B/Guangdong-XZ/37479/2021 were passaged in 10-day old specific pathogen free (SPF) eggs at 33°C for 3 days. All viruses were manipulated under BSL-2 conditions. All HA sequences of influenza B viruses used in this study are listed in [Table S5](#).

METHOD DETAILS

The isolation of plasmablasts

Blood samples were collected from the subject on day 7 after vaccination with the quadrivalent influenza virus split vaccine (Hualan Biological Vaccine Inc). Peripheral Blood Mononuclear Cells (PBMCs) were isolated by Ficoll-paque gradient assay (Lymphoprep™). The PBMCs were stained with a PB-labeled anti-human CD19 antibody (Biolegend), an APC-labeled anti-human CD27 antibody (Biolegend) and a PE-labeled anti-human CD38 antibody (Biolegend). Single CD19⁺CD27⁺CD38^{high} plasmablasts were sorted into 96-well plates using a FACSAria III (BD).^{19,20}

HA expression and purification

The ectodomain of HA used for screening were produced using HEK293T system. The HA_B/Florida/4/2006 (residues 1-529) and the HA_B/Colorado/06/2017 (residues 1-528) were constructed into a pcDNA3.1 vector with 6×His tag, and a trimerization domain to generate recombinant HA protein. The expression vectors were transiently transfected into HEK293T cells by polyethyleneimine (Polysciences). After 5 days of transfection, the culture supernatant was collected and the HA protein was purified using Ni-NTA Sepharose High-Performance (Cytiva). The HA were further concentrated using a 30K-cutoff concentrator (Millipore).

Insect cell system was used to express HA heads for crystallization because the proteins produced in this system have a less potential glycosylation, which was in favor for crystallization. The HA head region (residues 49-282 for HA_B/Florida/4/2006, residues 49-281 for HA_B/Colorado/06/2017, and residues 49-280 for HA_B/Guangdong-YX/120/2022), including wild-type and single point mutation, were cloned into the pFastBac dual vector with a gp67 signal peptide and a C-terminal 6×His tag for purification, respectively. The constructs were transformed into DH10Bac competent cells, from which bacmid was extracted and subsequently transfected into Sf9 cells using Cellfectin II Reagent (Invitrogen). The low-titered viruses were harvested and amplified to produce a high-titer virus stock, which were then used to infect Hi5 cells at a density of 2×10^6 cells/mL. 60 h post-infection, the supernatant of cell culture containing the secreted HA head were collected, concentrated and buffer-exchanged to HBS (10 mM HEPES, 150 mM NaCl). The HA head were captured using Ni-NTA Sepharose High-Performance (Cytiva) and eluted with 300 mM imidazole in HBS buffer. The eluted HA head were further concentrated using a 10K-cutoff concentrator (Millipore) and purified by gel filtration chromatography with the Superdex 200 column (Cytiva) pre-equilibrated with HBS buffer. Finally, the purified proteins were rapidly frozen in liquid nitrogen and stored at -80°C until further use.

Antibody expression and purification assay

The amino acid sequences of heavy and light chain variable regions of CR8033, CR8059, H1209, H1272, H2365 and C12G6 were downloaded from PDB. Antibodies were generated as previously described.^{20,27–29} Briefly, Immunoglobulin heavy and light chain variable genes from single cell sorted plasmablasts were amplified by reverse transcriptase polymerase chain reaction (RT-PCR) and nested PCR reactions then sequenced. The amplified heavy and light genes were cloned into IgG1 expression vectors. The supernatant was collected and purified using protein A agarose beads (Smart-lifesciences).

Enzyme-linked immunosorbent assay (ELISA)

Flat bottom 96-well plates (NUNC) were coated with 2 µg/mL purified recombinant protein or 8 HAU virus in PBS overnight at 4°C. After blocking with 3% BSA (GBCBio) in PBS, serially threefold dilutions of antibodies starting at 10 µg/mL were added to the plates and incubated for 1 h at 37°C. Plates were washed six times with PBST and then incubated with HRP (horseradish peroxidase)

conjugated goat anti-human IgG (Jackson). The plate was developed with Super Aquablu (Sigma-Aldrich) ELISA substrate. Absorbance was measured at 405 nm on a microplate spectrophotometer (MD SpectraMax ABS plus).

Hemagglutination inhibition assay (HI)

Hemagglutination inhibition assays were performed under the World Health Organization Manual on Animal Influenza Diagnosis and Surveillance. Serially diluted antibodies 1:2 starting at 50 µg/mL were mixed with an equal volume of 4 HAU virus and incubated for 30 min at room temperature. Chicken red blood cells (CRBCs) were added to the mixture and incubated at room temperature for 30 mins. Button formation was rated as evidence of hemagglutination inhibition.

Microneutralization assays (MN)

MN assays were carried out as described previously.^{28,30} Briefly, MDCK cells were cultured in DMEM with 10% FBS at 37°C with 5% CO₂. On the day of the experiment, MDCK cells in a 96-well plate were washed twice with PBS and incubated in DMEM supplemented with 1 µg/mL TPCK-treated Trypsin (Sigma-Aldrich). We prepared serial dilutions of each antibody, mixed them with an equal volume of virus, and incubated the mixture for 1 h at 37°C. After incubation, 100 µL of the mixture containing 100 TCID₅₀ (50% tissue culture infectious dose) of the virus was added to the MDCK cells and allowed to adsorb for 1 h. The viral supernatant was removed and replaced with DMEM supplemented with 1 µg/mL TPCK-treated Trypsin. Cells were cultured for 24 h at 37°C with 5% CO₂. The neutralization of the virus by the antibody is reflected by detecting the expression of NP in the cells. Absorbance was measured at 405 nm on a microplate spectrophotometer (MD SpectraMax ABS plus). The signal from uninfected wells were averaged to represent 100% inhibition. Virus infected wells without mAbs were averaged to represent 0% inhibition. Duplication wells were used to calculate the mean and SD of neutralization, and inhibitory concentration 50 (IC₅₀) was determined by a sigmoidal dose response curve.

Mouse challenge experiments

For the prophylactic experiments, 6-8-week-old female BALB/c mice were injected intraperitoneally (i.p.) with 200 µL of antibodies (n = 5 per group). After 1 day treatment, the mice were anesthetized using Avertin (Tribromoethanol, TBE) and intranasally infected with 10 LD₅₀ (50% lethal dose) of MA-B/Colorado/06/2017, MA-B/Florida/4/2006 and MA-B/Guangdong-YX/120/2022 in a 30 µL inoculum. In therapeutic studies, mice received 200 µL of antibodies i.p. 24 h after 10 LD₅₀ virus intranasally infection. All mice were monitored daily for mortality and morbidity, and body weight was measured until day 14 post-infection. The endpoint was defined as mice achieving a weight reduction of more than 25%. Lung viral titers after challenge with MA-B/Florida/4/2006 (Y) and MA-B/Colorado/06/2017 (V). Right lung tissue was collected on day 4 post-infection for titer determination. The prophylaxis group received a single 5 mg/kg antibody dose 24 h before infection, whereas the therapy group was treated with 10 mg/kg antibody 24 h after infection. The viral load in lung homogenates was quantified to determine the pulmonary viral titer in mice.

Sequence conservation analysis

A total of 72604 full-length IBV(B/Victoria and B/Yamagata)HA protein sequences were downloaded from Global Initiative for Sharing Avian Influenza Data (GISAID; <https://gisaid.org>). HA-B/Florida/4/2006(Y) and HA-B/Colorado/06/2017(V) amino acid sequences were chosen as references in multiple sequences alignment using mafft tool, respectively. Sequences (<0.05%) with insertions/deletions ratio greater than 10% were removed from alignment. Conservation of filtered sequences were showed with seqlogo plot using ggseqlogo package.

Viral immune escape selection

B/Colorado/06/2017 and B/Florida/04/2006 were serially passaged in MDCK cells under antibody pressure. Antibodies were two-fold serially diluted starting 10 µg/mL and co-incubated with 50,000 TCID₅₀ of virus for 1 h. The antibody-virus mixture was then added to MDCK cells and incubated at 37 °C for 1 h. After removal of the inoculum, the cells were washed twice with PBS and overlaid with maintenance medium containing 1 µg/mL antibody. Cytopathic effect (CPE) and hemagglutination unit (HAU) titers were monitored at 72 h post-infection.^{12,13} Viral RNA was extracted from culture supernatants, and the HA gene was amplified by RT-PCR. Amplicons were subjected to PCR-free library preparation and sequenced using paired-end 250 bp (PE250) Illumina sequencing. Sequencing reads were quality-filtered using FastQC and Trim Galore to remove adapters and low-quality bases. Clean reads were aligned to strain-matched HA reference sequences using BWA, and the alignments were processed with SAMtools.³¹ Single nucleotide variants were identified using LoFreq³² and functionally annotated with snpEff.³³

Phylogenetic tree of IBV HA

IBV HA nucleotide sequences in this study were analyzed using Nextclade. Sequences were aligned and placed onto the reference phylogeny using the Influenza B Victoria HA dataset (dataset name: nextstrain/flu/vic/ha/KX058884). Phylogenetic relationships were inferred based on the resulting placement, and sequences were annotated and colored according to the Legacy clade scheme.³⁴

ADCC assay

MDCK cells were seeded at 2 × 10⁴ cells per well in 96-well plates and cultured in DMEM containing 10% FBS at 37°C with 5% CO₂. On the day of the experiment, the cells were washed twice with PBS, then inoculated with 100 µL of influenza virus at an MOI of 5 for 1 h. After removal of the viral inoculum, the medium was replaced with DMEM supplemented with 1 µg/mL TPCK-treated Trypsin, and

the cells were incubated for an additional 24 h under the same conditions. Antibodies were diluted in RPMI-1640 containing low-IgG serum (ABW) in five-fold serial dilutions starting from 20 $\mu\text{g/mL}$. Following two PBS washes, 25 μL of diluted antibody was added to each well and incubated for 1 h. Subsequently, 25 μL of Jurkat-Lucia™ NFAT-CD16 cells (jktl-nfat-cd16) ($6 \times 10^6/\text{mL}$) were added per well, and the plates were incubated at 37°C with 5% CO_2 for 24 h. Luciferase activity was quantified using QUANTI-Luc™ 4 Lucia/Gaussia (rep-q1c4lg1) reagent.

Biolayer interferometry binding assay (BLI)

Binding assays were performed by biolayer interferometry (BLI) using an Octet R8 instrument (Sartorius). Briefly, the antibodies diluted at 5 $\mu\text{g/mL}$ in PBS (pH 7.4, added 0.02% Tween 20) was loaded onto protein A biosensors and incubated with 25 nM, 12.5 nM, 6.25 nM, 3.125 nM and 1.5625 nM of HA proteins for 120 s or 300 s. Following with 1000s disassociation in assay buffer, sensors were regenerated in 10 mM glycine (pH=1.5) for 30s per cycle. For estimating the K_D , a 1:1 binding model was used. The fitting data was analyzed by GraphPad Prism 8.0.

Crystallization of the HA-antibody complexes and data collection

The purified HA head of B/Colorado/06/2017 or B/Guangdong-YX/120/2022 was mixed with CAV-CH76 Fab at a 1:1 molar ratio, while the HA head of B/Florida/4/2006 or B/Guangdong-YX/120/2022 was mixed with CAV-CF22 Fab under the same condition. Each resulting complex was purified and concentrated to ~ 10 mg/mL in HBS buffer (10 mM HEPES, pH 7.2, 150 mM NaCl). Crystals of all complexes were successfully cultivated at 16°C in sitting drops by mixing 0.15 μL protein with 0.15 μL reservoir solution. The HA head of B/Colorado/06/2017-CAV-CH76 Fab complex crystallized successfully in a solution containing 0.15 M DL-Malic acid (pH 7.0) and 20% w/v Polyethylene glycol 3350, while HA head of B/Guangdong-YX/120/2022-CAV-CH76 Fab complex crystallized in a solution containing 0.2 M Magnesium chloride hexahydrate, 0.1 M Tris (pH 8.5), 25% w/v Polyethylene glycol 3350. Meanwhile, the HA of B/Florida/4/2006-CAV-CF22 Fab complex formed crystals in a solution of 1 M MES (pH 6.0), 50% w/v PEG8000, and 1 M $\text{Zn}(\text{OAc})_2$. The HA head of B/Guangdong-YX/120/2022-CAV-CF22 Fab complex crystallized successfully in a solution containing 0.2 M Lithium sulfate monohydrate, 0.1 M BIS-TRIS (pH 6.5), 25% w/v Polyethylene glycol 3350. Following crystallization, the crystals were flash-frozen in liquid nitrogen. Diffraction data was collected at 100 K and at a wavelength of 0.97918Å utilizing the BL18U1 beam line for HA-head (B/Colorado/06/2017)-CAV-CH76 Fab complex and BL02U1 beam line for the other three complex at the Shanghai Synchrotron Radiation Facility. The diffraction data were processed with HKL2000 software³⁵ and the detailed data-processing statistics are summarized in [Table S1](#).

Structure determination and refinement

The structure determination was performed using the molecular replacement method implement in PHASER within the CCP4 suite.³⁶ The search models included the HA head, CAV-CH76 Fab and CAV-CF22 Fab, which were predicted by AlphaFold2.³⁷ Subsequently, the progression of model construction and refinement was performed using COOT and PHENIX, respectively.^{38,39} The final Ramachandran statistics for the HA-head_B/Colorado/06/2017-CAV-CH76 Fab complex showed 96.80% favored, 3.20% allowed and 0.00% outlier, and the HA-head_B/Guangdong-YX/120/2022-CAV-CH76 Fab complex showed 95.43% favored, 4.11% allowed and 0.46% outlier. For the complexes with CAV-CF22 Fab, the HA-head_B/Florida/4/2006 showed 93.30% favored, 6.59% allowed and 0.11% outlier, while the HA-head_B/Guangdong-YX/120/2022 structure showed 94.70% favored, 5.00% allowed and 0.30% outlier. Detailed statistics pertaining to the refinement of the structure are summarized in [Table S1](#). The visualization of all structural representations were executed utilizing chimera X.⁴⁰

QUANTIFICATION AND STATISTICAL ANALYSIS

All statistical analysis was performed by Prism software (GraphPad Prism Version 8.0). The 50% inhibitory concentration (IC_{50}) values by neutralization assays were calculated by nonlinear regression analysis using the GraphPad Prism software. Weight changes were determined compared to the baseline weight of each mouse measured on day 0. $n=5$ mice per group. The endpoint was defined as mice achieving a weight reduction of more than 25%. The statistical significance of lung viral titers was assessed using the Mann-Whitney test: * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$.